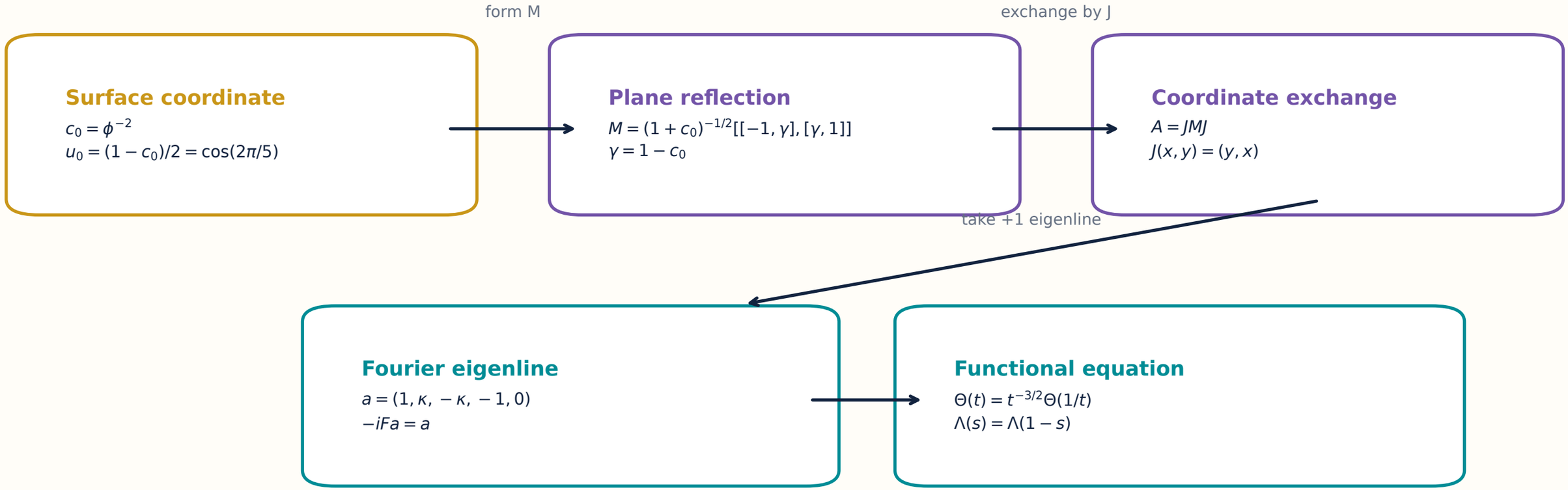
Figure 01. One scalar coordinate, an explicit involution, and its Fourier consequence



Distinct CM branch: i is fixed by $S : \tau \mapsto -1/\tau$ and selects $\kappa = r(i)$.
$\tau_{163} = (1 + i\sqrt{163})/2$ instead supplies $j(\tau_{163}) = -640320^3$ and a tiny negative nome.

Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue